



```
# 1 Write a program that takes two numbers and prints the result of all 7 arithmetic operations (+, -, *, /, //, %, **).
a_ankit=2
b_ankit=7
print(f'The sum of {a_ankit} and {b_ankit} is {a_ankit+b_ankit}')
print(f'The difference of {a_ankit} and {b_ankit} is {a_ankit-b_ankit}')
print(f'The multiplication of {a_ankit} and {b_ankit} is {a_ankit*b_ankit}')
print(f'The division of {a_ankit} and {b_ankit} is {a_ankit/b_ankit}')
print(f'The floor division of {a_ankit} and {b_ankit} is {a_ankit//b_ankit}')
print(f'The modulator division of {a_ankit} and {b_ankit} is {a_ankit%b_ankit}')
print(f'The exponential of {a_ankit} and {b_ankit} is {a_ankit**b_ankit}')
```

```
*** The sum of 2 and 7 is 9
The difference of 2 and 7 is -5
The multiplication of 2 and 7 is 14
The division of 2 and 7 is 0.2857142857142857
The floor division of 2 and 7 is 0
The modulator division of 2 and 7 is 2
The exponential of 2 and 7 is 128
```

▶ # 2 Create a tip calculator: take a bill amount and tip percentage, calculate the tip and total.

```
amount_ankit=455
tip_percent=20
tip_ankit=20/100*amount_ankit
total_ankit=amount_ankit+tip_ankit
print(f'The total is {total_ankit} with tip {tip_ankit}')
```

*** The total is 546.0 with tip 91.0



```
# 3 Check if a number is even or odd using the modulus operator (%).  
num_ankit=34  
if num_ankit%2==0:  
    print(f'{num_ankit} is even')  
else:  
    print(f'{num_ankit} is odd')
```

```
*** 34 is even
```



```
# 4 Ask for the user's name and age, then print: "Hello [name], you are [age] years old."  
name_ankit=input('Enter your Name')  
age_ankit=input('Enter your age')  
print(f"Hello {name_ankit}, you are {age_ankit} years old.")
```

```
*** Enter your Nameankit  
Enter your age20  
Hello ankit, you are 20 years old.
```

▶ # 5 Ask for a temperature in Celsius and convert to Fahrenheit using $F = C * 9/5 + 32$. Display the result with 1 decimal place

```
temp_ankit=int(input('Enter the temperature in celcius'))  
f_ankit= temp_ankit * 9/5 + 32  
print(f'The temperature in fahrenheit is {f_ankit:.1f}')
```

```
*** Enter the temperature in celcius32  
The temperature in fahrenheit is 89.6
```



```
# 6 Create a formatted receipt: ask for 3 items and their prices, display each item and the total.
i1_ankit=input('Enter the item 1 ')
p1_ankit=int(input('Enter the price of item 1 '))

i2_ankit=input('Enter the item 2 ')
p2_ankit=int(input('Enter the price of item 2 '))

i3_ankit=input('Enter the item 3 ')
p3_ankit=int(input('Enter the price of item 3 '))

t_ankit = p1_ankit+p2_ankit+p3_ankit
print(f'{i1_ankit} has price {p1_ankit}')
print(f'{i2_ankit} has price {p2_ankit}')
print(f'{i3_ankit} has price {p3_ankit}')
print(f'The total price is {t_ankit}')
```

```
*** Enter the item 1 pen
Enter the price of item 1 20
Enter the item 2 pencil
Enter the price of item 2 10
Enter the item 3 eraser
Enter the price of item 3 5
pen has price 20
pencil has price 10
eraser has price 5
The total price is 35
```



```
# 7 Ask for a username and password. Print "Access granted" only if both are correct.
```

```
u_ankit=input('Enter your username')
```

```
p_ankit=input('Enter your password')
```

```
if u_ankit== 'user' and p_ankit == 'password':
```

```
    print('Access granted')
```

```
else:
```

```
    print('Access denied')
```

```
# 8 Check if a number is between 1 and 100 (inclusive) using and.
```

```
number_ankit=75
```

```
if number_ankit < 100 and number_ankit > 1:
```

```
    print(f'{number_ankit} is between 1 and 100')
```

```
# 9 Write a program that checks if a year is a leap year (divisible by 4 and not 100, or divisible by 400).
```

```
y_ankit=2026
```

```
if y_ankit%4 == 0 and y_ankit !=100 or y_ankit%400 == 0:
```

```
    print(f'{y_ankit} is a leap year')
```

```
*** Enter your usernameankit
Enter your passwordankit123
Access denied
75 is between 1 and 100
```



```
# 10 Ask the user for a number. Print whether it is positive, negative, or zero.  
ankit_num=int(input('Enter the number'))  
if ankit_num < 0 :  
    print(f'{ankit_num} is negative number')  
elif ankit_num > 0:  
    print(f'{ankit_num} is positive')  
else:  
    print(f'{ankit_num} is a zero')
```

```
*** Enter the number2  
2 is positive
```




```
# 11 Write a grading system: take a score (0-100) and print the letter grade (A, B, C, D, or F).
s_ankit=int(input('Enter the score must be between 1 and 100'))
if s_ankit >100 or s_ankit < 0:
    print(f'Invalid Score')
elif s_ankit >= 90:
    print("A")
elif s_ankit >= 80:
    print("B")
elif s_ankit >= 70:
    print("C")
elif s_ankit >= 60:
    print("D")
else:
    print("F")
```

```
*** Enter the score must be between 1 and 10067
D
```

```
# 12 Ask for an age and print the ticket price: child (under 12) = $5, adult (12-64) = $10, senior (65+)
ankit_age=int(input('Enter your age'))
if ankit_age < 0 :
    print('Invalid Age')
elif ankit_age < 12:
    print(f'The ticket price is $5')
elif 12 < ankit_age <64:
    print(f'The ticket price is $10')
else:
    print(f'The ticket price is $7')

# 13 Print the multiplication table for any number (1 to 10) using a for loop.
for i in range(1,11):
    print(f'5 x {i} = {5*i}')
```

```
Enter your age20
The ticket price is $10
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
```



```
# 14 Loop through a list of 5 names and print each with its position (use enumerate).
```

```
names_ankit=['Ankit','Rohit','Saurabh','Ramesh','Rakesh']
```

```
for index, name in enumerate(names_ankit):
```

```
    print(f'Position {index+1}: {name}')
```

```
# 15 Calculate the sum of all even numbers from 1 to 50 using range().
```

```
sum_even_ankit=0
```

```
for i in range(1,51):
```

```
    if i%2==0:
```

```
        sum_even_ankit+=i
```

```
print(f'The sum of all even numbers from 1 to 50 is {sum_even_ankit}')
```

```
***
```

```
Position 1: Ankit
```

```
Position 2: Rohit
```

```
Position 3: Saurabh
```

```
Position 4: Ramesh
```

```
Position 5: Rakesh
```

```
The sum of all even numbers from 1 to 50 is 650
```



```
# 16 Write a countdown from 10 to 1 using a while loop, then print "Liftoff!"  
print("Countdown:")  
i = 10  
while i >= 1:  
    print(i)  
    i -= 1  
print("Liftoff!")
```

*** Countdown:

10

9

8

7

6

5

4

3

2

1

Liftoff!



```
# 17 | keep asking the user for a password until they enter "secret123".  
password = ""  
while password != "secret123":  
    password = input("Please enter the password: ")  
    if password != "secret123":  
        print("Incorrect password. Try again.")  
print("Password accepted!")
```

```
*** Please enter the password: secret123  
Password accepted!
```



```
# 18 | create a simple number guessing game: pick a number, let the user guess until correct.
import random
secret_number = random.randint(1, 10) # Pick a number between 1 and 10
guess = 0

print("Welcome to the Number Guessing Game!")
print("I've picked a number between 1 and 10. Can you guess it?")

while guess != secret_number:
    try:
        guess = int(input("Enter your guess: "))
        if guess < secret_number:
            print("Too low! Try again.")
        elif guess > secret_number:
            print("Too high! Try again.")
    except ValueError:
        print("Invalid input. Please enter a number.")

print(f"Congratulations! You guessed the number {secret_number}!")
```

```
*** Welcome to the Number Guessing Game!
I've picked a number between 1 and 10. Can you guess it?
Enter your guess: 1
Too low! Try again.
Enter your guess: 4
Too high! Try again.
Enter your guess: 3
Congratulations! You guessed the number 3!
```

19 Write a function is_even(number) that returns True if the number is even, False otherwise.

```
def is_even(number):  
    if number%2==0:  
        return True  
    else:  
        return False  
is_even(2)
```

True

20 Create a function calculate_area(length, width) that returns the area. Use it to calculate areas of 3 different rectangle

```
def calculate_area(length,width):  
    area = length * width  
    return area  
calculate_area(2,4)  
calculate_area(10,4)  
calculate_area(2,8)
```

```
# 21 Write a function create_receipt(item, price, quantity) that returns a formatted string with the total
def create_receipt(item, price, quantity):
    total=price*quantity
    print(f'The total price of {quantity} {item} is {total}')
create_receipt('Pen', 20, 5)
```

The total price of 5 Pen is 100